

Access Queries, Forms & Reports

Lesson 30 — Objects, Queries and Reporting

JKSSB Computer Awareness

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The Four Main Access Objects

Object	Job
Table	Stores the actual data in rows and columns
Query	Asks a question and retrieves selected data
Form	Provides a screen to enter and view data
Report	Presents data in a printable layout

Data lives in tables; queries, forms and reports work on that data.

The Object Sequence

Table → **Query** → **Form** → **Report**

- ▶ **Table**: the store where the data actually lives.
- ▶ **Query**: asks a question and pulls the needed rows together.
- ▶ **Form**: a screen for entering and viewing data.
- ▶ **Report**: a formatted, printable presentation of the data.

The one idea

Data is stored once, in a table; the query selects it, the form enters it, and the report prints it.

Tables Revisited: The Source of Data

- ▶ A **table** stores data in **fields** (columns) and **records** (rows).
- ▶ Every other object draws its data from a **table** or a **query**.
- ▶ A form or a report built on a table shows that table's data; built on a query, it shows only the **selected** rows.
- ▶ The table is therefore the **foundation** of the database.

Remember

No data is stored in a query, form or report — they only present what a table holds.

What Is a Query?

- ▶ A **query** is a question you ask of the database.
- ▶ It **retrieves** data from one or more tables, usually by a condition.
- ▶ A query does not normally store new data; it shows a result **set** drawn from the tables.
- ▶ Queries can also **modify** data — these are called **action queries**.

Definition

A query selects, filters, or acts on records according to the rules you set.

The Select Query

- ▶ A **Select query retrieves** records that satisfy a **condition** (a criterion).
- ▶ It is the **most common** query type in Access.
- ▶ Example: show every student whose Marks are greater than 60.
- ▶ The condition decides **which rows appear**; the fields decide **which columns appear**.

Exam point (PYQ-39)

A query that retrieves data on the basis of a condition is a **Select Query**.

Building a Query: Design View and Criteria

- ▶ Queries are built in **Query Design** view (or typed in **SQL** view).
- ▶ The design grid is called **QBE — Query By Example**.
- ▶ Each column holds a **field**; the **Criteria** row holds the condition.
- ▶ Example criterion: >60 on the Marks field shows only rows above 60.

Worked example

For a Students table, put Marks in the field row and >60 in the Criteria row; the query returns only students who scored more than 60.

Action Queries: Update, Append, Delete

Query	What it does to the data
Update	Changes existing values in records
Append	Adds records from one table to another
Delete	Removes records that match a condition

Update, Append and Delete all change the stored data; that is why they are called action queries.

Select vs Action Queries (Near-Neighbour)

Point	Select query	Action query
What it does	Retrieves data on a condition	Modifies stored data
Changes data?	No	Yes
Examples	Select	Update, Append, Delete

Trap alert (TRAP-31)

A **Select** query only **retrieves**; to modify data use an **Update**, **Append**, or **Delete** query.

Lookup Fields

- ▶ A **lookup** field lets the user **pick a value from a list** instead of typing it.
- ▶ The list usually comes from a **table** or a **query**, such as a list of department names.
- ▶ A lookup is often used for a **foreign key**, so the entry matches an existing value.
- ▶ Benefit: it **reduces typing errors** and keeps values consistent.

Keep it clear

A lookup field *chooses* a value from a list; it does not create a new table or a new query.

Basic SQL Awareness

- ▶ **SQL** stands for **Structured Query Language**.
- ▶ It is the standard language for working with relational databases; Access runs SQL behind the scenes.
- ▶ A typical retrieval statement is `SELECT fields FROM table WHERE condition;`.
- ▶ `SELECT` retrieves; `FROM` names the table; `WHERE` gives the condition.

Example

`SELECT Name FROM Students WHERE Marks > 60;` returns the Name of every student with marks above 60.

Forms: Entering and Viewing Data

- ▶ A **form** is an object that provides a **screen** — a user interface — for entering, viewing, and editing data.
- ▶ A form is usually **based on a table or a query**.
- ▶ It shows one record at a time (or a few), with clearly labelled fields.
- ▶ Forms make data entry **easier and safer** than typing directly into a table grid.

Exam point

A form is for **on-screen** work with data — entering and viewing it.

Reports: Printable Output

- ▶ A **report** presents data in a **formatted, printable** layout.
- ▶ It can **group** records and show **totals and summaries**.
- ▶ A report is **read-only**: it is meant for **printing and sharing**, not for editing data.
- ▶ A report can be based on a table or a query.

Definition

A report is the print-ready view of your data — the output object.

Form vs Report (Near-Neighbour)

Point	Form	Report
Purpose	Enter and view data on screen	Print and present data
Data change	Allows editing	Read-only
Typical use	Data-entry screen	Printed summary

Form = screen for data entry; Report = printed output. Do not swap them.

Worked Example: A Students Database

- ▶ **Table:** Students with RollNo, Name, Marks.
- ▶ **Query:** select students with Marks greater than 60 (a Select query).
- ▶ **Form:** a screen to enter a new student's details.
- ▶ **Report:** a printable list of all students who passed.

The full sequence

Table (stores the data) → Query (selects it) → Form (enters it) → Report (prints it).

Trap Alert: Objects and Query Types

- ▶ A **Select** query **retrieves**; **Update**, **Append** and **Delete** queries **modify** (TRAP-31).
- ▶ Data is stored in a **table**, **not** in a query, form or report.
- ▶ A **form** is for **on-screen entry**; a **report** is a **printable** output.
- ▶ **SQL** = Structured Query Language.
- ▶ A **lookup** field picks a value **from a list**.

Lesson 30: Recall

- ▶ The four main objects: **Table** (stores), **Query** (retrieves), **Form** (enters on screen), **Report** (prints).
- ▶ The sequence is **Table** → **Query** → **Form** → **Report**.
- ▶ A **Select query** retrieves data on a condition; **action queries** (Update, Append, Delete) modify data.
- ▶ Queries are built in **QBE** (Query By Example) design view with a **Criteria** row.
- ▶ A **lookup** field chooses a value from a list.
- ▶ **SQL** (Structured Query Language) uses `SELECT ... FROM ... WHERE`.